

- 2 Fig. 2.1 is a transmission electron micrograph of a section through a blood vessel.

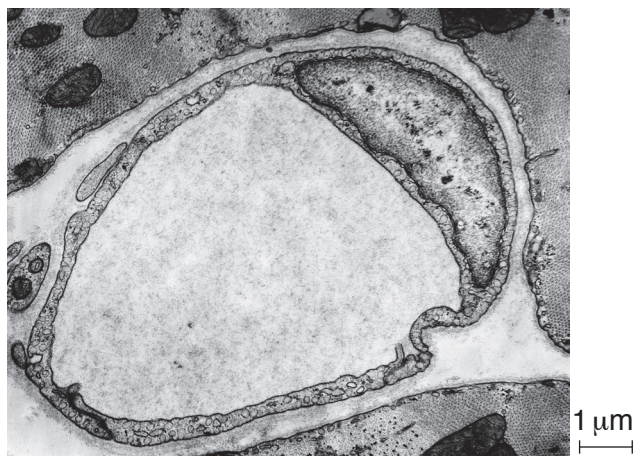


Fig. 2.1

- (a) State the type of blood vessel shown in Fig. 2.1 **and** give **two** reasons for your choice.

type of blood vessel

reason 1

.....

reason 2

.....[3]

The graph shows the blood pressure in the pulmonary artery (solid line) and the right ventricle (dashed line) over a period of 0.8 seconds. The y-axis represents blood pressure in kPa, ranging from 0 to 4. The x-axis represents time in seconds, ranging from 0.0 to 0.8. The pulmonary artery pressure starts at approximately 1.7 kPa, remains constant until 0.1 s, then rises to a peak of about 3.4 kPa at 0.3 s, before gradually declining to about 1.7 kPa by 0.8 s. The right ventricle pressure starts at 0 kPa, remains at 0 until 0.1 s, then rises sharply to a peak of about 3.2 kPa at 0.25 s, and then falls back to 0 kPa by 0.7 s.

time/s	pulmonary artery /kPa	right ventricle /kPa
0.0	1.7	0.0
0.1	1.7	0.0
0.2	3.0	3.2
0.3	3.4	2.8
0.4	2.8	2.0
0.5	2.1	0.5
0.6	2.0	0.2
0.7	1.8	0.0
0.8	1.7	0.0

h:

-[4]